

Intelligent Decentralized Disaster Response System using Adaptive AI and Mesh Networking

Abstract—Natural disasters often damage communication infrastructure, making rescue operations slow and inefficient during emergency situations. Most existing disaster management systems depend heavily on internet connectivity and centralized communication networks, which may fail during floods, earthquakes, cyclones, and other large-scale disasters. To address these challenges, this paper presents an Intelligent Decentralized Disaster Response System using Adaptive Artificial Intelligence and Mesh Networking. The proposed system enables communication even without internet access by using Bluetooth Low Energy (BLE), Wi-Fi Direct, and multi-hop mesh networking between nearby mobile devices. An AI-based adaptive routing mechanism is used to identify reliable communication paths by considering factors such as signal strength, battery level, device availability, and network congestion. The system also incorporates victim severity analysis and rescue prioritization to support faster emergency response. In addition, graph-based routing algorithms such as Dijkstra and A* are used to improve message delivery and optimize communication efficiency. A real-time monitoring dashboard is integrated to assist rescue teams in coordination and resource allocation. Experimental analysis demonstrates improved communication reliability, routing efficiency, and response performance compared to traditional centralized systems. The proposed framework provides a scalable, reliable, and intelligent solution for disaster response communication in infrastructure-failure scenarios.

Keywords—Disaster Response, Mesh Networking, Artificial Intelligence, Wireless Mesh Networks, BLE Communication, Adaptive Routing, Emergency Communication, Decentralized Systems, Rescue Prioritization, Disaster Management.

I. INTRODUCTION

Natural disasters such as floods, earthquakes, cyclones, and landslides often damage communication infrastructure, making rescue operations difficult and delaying emergency response activities. In many disaster situations, mobile towers, internet services, and centralized communication systems stop functioning due to network failure or physical damage. As a result, affected people may not be able to send emergency alerts or share their location with rescue teams. Reliable communication therefore becomes one of the most important requirements during disaster management.

Most existing disaster response systems depend heavily on internet connectivity and centralized communication networks. Although these systems work effectively under normal conditions, they become unreliable when communication infrastructure is damaged. In addition, many existing systems do not support offline communication or intelligent routing, which can lead to delays, communication gaps, and poor coordination during emergencies.

Recent developments in Wireless Mesh Networks (WMNs), Bluetooth Low Energy (BLE), Wi-Fi Direct, and Artificial Intelligence (AI) have made it possible to build decentralized communication systems that can operate even without internet access. In mesh networking, nearby devices communicate directly with each other and forward messages across multiple devices using multi-hop communication. AI-based adaptive routing techniques can further improve communication reliability by selecting efficient communication paths based on factors such as signal strength, battery level, and network conditions.

This paper proposes an Intelligent Decentralized Disaster

Response System using Adaptive AI and Mesh Networking for reliable emergency communication during disaster situations. The proposed system combines decentralized mesh networking, adaptive routing, rescue prioritization, and real-time monitoring to improve communication reliability, coordination efficiency, and overall disaster response performance.

II. LITERATURE SURVEY

Several researchers have focused on improving disaster communication systems using Wireless Mesh Networks (WMNs), Artificial Intelligence (AI), IoT, and decentralized networking technologies. Most studies aim to improve communication reliability, routing efficiency, and rescue coordination during emergency situations.

Alhammad et al. [1] discussed the role of advanced wireless and aerial communication technologies in disaster management. Luo et al. [2] proposed UAV-assisted emergency communication networks to improve connectivity during disaster recovery situations. Nishiyama et al. [3] introduced smartphone-based multi-hop communication for device-to-device emergency networking.

Abidde et al. [4] presented an AI-based WMN framework for adaptive routing and improved communication efficiency. Tran et al. [5] developed UAV relay-assisted emergency communication methods for IoT-based disaster networks. Álvarez et al. [6] proposed Bluemergency, a Bluetooth mesh-based offline communication system for post-disaster communication.

Other studies [7], [8] explored AI and IoT technologies for disaster prediction, emergency communication, and rescue coordination. Rohde et al. [9] and Zeng et al. [10] investigated UAV-assisted and wireless communication approaches to improve network coverage during emergency situations. Akyildiz et al. [11] highlighted the decentralized, scalable, and self-healing features of Wireless Mesh Networks for disaster communication.

Ortigoso et al. [12] proposed a resilient smart city communication architecture using Wi-Fi HaLow mesh networks for post-disaster connectivity. Ashraf et al. [13] developed WiMesh, a low-cost mesh networking solution for disaster-affected regions. Asplund et al. [14] discussed IoT security considerations in critical emergency communication systems, while Liu et al. [15] proposed scalable routing techniques for city-scale Wi-Fi disaster recovery networks.

Although existing disaster communication systems improve communication reliability and routing efficiency, most focus only on individual functionalities such as offline communication, routing, or emergency monitoring. Very few systems provide an integrated framework that combines decentralized mesh networking, adaptive AI-based routing, rescue prioritization, and real-time monitoring within a single platform. In addition, many existing approaches do not consider multiple real-time factors such as battery level, signal strength, node availability, and network congestion

while selecting communication paths. These limitations can reduce communication stability and rescue coordination during disaster situations. The proposed system addresses these challenges through an intelligent decentralized disaster response framework.

III. PROBLEM STATEMENT

Natural disasters such as floods, earthquakes, cyclones, and landslides often cause serious damage to communication infrastructure, including mobile towers, internet services, and centralized networks. During such situations, affected people may not be able to send emergency alerts or share their location with rescue teams, leading to delays in rescue operations and poor coordination. Most existing disaster response systems rely heavily on internet connectivity and centralized communication mechanisms, which may become unavailable or unstable during disasters. In addition, many current systems do not support reliable offline communication, intelligent routing, rescue prioritization, or real-time coordination between affected individuals and emergency responders. These limitations reduce the overall efficiency of disaster management and emergency response operations. Therefore, there is a need for an intelligent and decentralized disaster response system that can provide reliable communication, adaptive routing, and efficient rescue coordination even in situations where conventional communication infrastructure is unavailable.

IV. PROPOSED METHODOLOGY

A. Decentralized Mesh Communication

The proposed system uses a decentralized mesh networking approach to maintain communication during disaster situations where internet services and mobile networks may become unavailable. Instead of depending on centralized servers or communication towers, nearby mobile devices communicate directly with each other using Bluetooth Low Energy (BLE) and Wi-Fi Direct technologies. This allows devices within the affected area to form a temporary mesh network for emergency communication.

In this network, every mobile device acts as a mesh node capable of sending, receiving, storing, and forwarding emergency messages to neighboring devices. Messages are transferred through multi-hop communication, where one device forwards the message to another until it reaches rescue teams or connected monitoring systems. This helps maintain communication even when conventional communication infrastructure is damaged or completely unavailable.

The decentralized architecture also improves communication reliability during emergencies. If one communication path becomes unavailable because of weak connectivity, device movement, or low battery, the system automatically identifies an alternative route through nearby devices. This self-organizing behavior helps maintain stable communication in dynamic disaster environments.

By using BLE and Wi-Fi Direct technologies, the system supports low-power and infrastructure-independent communication between nearby devices. As a result, the proposed mesh communication model provides a reliable and scalable solution for emergency communication and disaster response coordination.

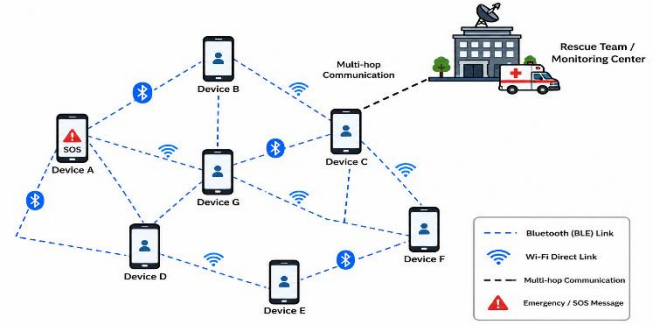


Fig. 1. Decentralized mesh communication between nearby devices during disaster situations.

Fig. 1 shows how nearby devices communicate through a decentralized mesh network during disaster situations.

B. Adaptive AI-Based Routing

The proposed system uses an adaptive AI-based routing mechanism to improve communication efficiency within the mesh network. During disaster situations, network conditions may change frequently because of signal fluctuations, device movement, battery limitations, or network congestion. To handle these challenges, the routing mechanism continuously selects efficient communication paths based on real-time network conditions.

Before forwarding an emergency message, the system evaluates nearby nodes using parameters such as signal strength, battery level, node availability, and communication traffic. Based on these factors, the system selects the most suitable route for message transmission. This helps reduce communication delays and improves successful message delivery during emergencies.

The routing score is calculated using:

$$\text{Routing Score} = \alpha(\text{Signal Strength}) + \beta(\text{Battery Level}) - \gamma(\text{Network Congestion}) + \delta(\text{Node Availability})$$

where α , β , γ , and δ represent weighting factors assigned to different routing parameters. Higher routing scores indicate more reliable communication paths within the network.

If a node becomes unavailable because of low battery or weak signal strength, the system automatically reroutes the message through another available path. In addition, graph-based algorithms such as Dijkstra and A* are used to identify shorter and more efficient routes for communication.

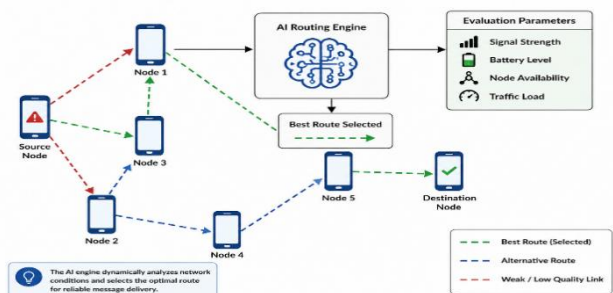


Fig. 2. Adaptive AI-based routing process within the mesh network.

Fig. 2 illustrates the adaptive routing mechanism used to select reliable communication paths within the mesh network.

C. Rescue Prioritization Mechanism

During disaster situations, rescue teams often receive multiple emergency requests simultaneously. To improve rescue coordination, the proposed system includes a rescue prioritization mechanism that helps identify critical cases more efficiently.

When a user sends an emergency alert, the system analyzes factors such as injury severity, environmental risk, victim vulnerability, and accessibility. Based on this information, emergency requests are categorized into different priority levels so that critical cases can be handled before less severe situations.

The Rescue Priority Index (RPI) is calculated using:

$$RPI = w_1(\text{Injury Severity}) + w_2(\text{Environmental Risk}) + w_3(\text{Victim Vulnerability}) - w_4(\text{Accessibility})$$

where w_1 , w_2 , w_3 , and w_4 represent weighting factors assigned to different emergency conditions. Higher RPI values indicate higher rescue priority during disaster situations.

High-priority requests are highlighted on the monitoring dashboard, allowing rescue teams to quickly identify people who require immediate assistance. This improves rescue coordination and emergency response efficiency.

D. Real-Time Monitoring and Coordination

The proposed system includes a real-time monitoring and coordination module to help rescue teams manage emergency situations more effectively. A monitoring dashboard is used to display emergency alerts, communication activity, rescue status, and resource information in real time. This allows rescue teams to continuously monitor disaster situations and coordinate rescue operations more efficiently.

Whenever a user sends an emergency request, the information is forwarded through the mesh network and displayed on the dashboard along with details such as location, type of emergency, priority level, and communication status. This helps rescue teams quickly identify critical situations and respond without unnecessary delay.

The dashboard also supports resource management by helping rescue teams monitor the availability of medical support, transportation, food supplies, and rescue personnel. Emergency requests can be tracked based on their current status, which improves coordination between rescue teams and affected individuals during disaster situations.

In addition, the monitoring system provides a clear view of communication activity within the decentralized mesh network. This helps maintain communication continuity and improves overall disaster response management. Overall, the real-time monitoring and coordination module improves rescue efficiency, communication management, and emergency response operations.

V. PROPOSED ARCHITECTURE

A. Overview of the Architecture

The proposed architecture combines mesh networking, adaptive routing, rescue prioritization, and real-time monitoring to support emergency communication during disaster situations. The system is designed to operate even when conventional communication infrastructure becomes unavailable.

B. Mesh Networking Layer

The mesh networking layer enables direct communication between nearby devices using Bluetooth Low Energy (BLE) and Wi-Fi Direct. Emergency messages are forwarded through multiple devices, allowing communication to continue without internet connectivity.

C. Adaptive Routing Layer

The adaptive routing layer selects suitable communication paths by considering factors such as signal strength, battery level, node availability, and network congestion. Dijkstra and A* algorithms are used to improve routing efficiency and message delivery.

D. Rescue Coordination Layer

The rescue coordination layer processes incoming emergency requests and assigns priority levels based on urgency and severity. This helps rescue teams identify critical situations and respond more effectively.

E. Monitoring Dashboard

The monitoring dashboard provides real-time information about emergency alerts, communication status, rescue activities, and available resources. It helps rescue teams monitor ongoing situations and coordinate response operations efficiently.

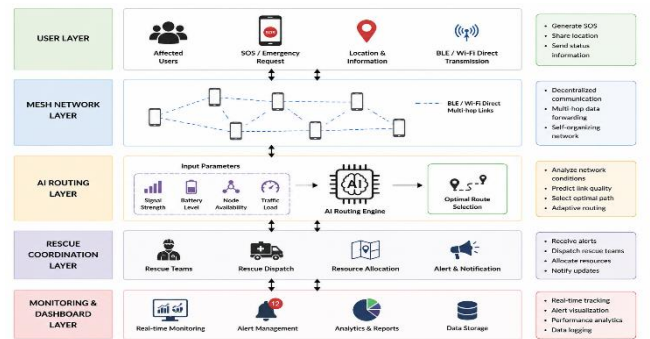


Fig. 3. Proposed architecture of the decentralized disaster response system.

Fig. 3 presents the overall architecture of the proposed disaster response system and its major components.

F. Contributions of the Proposed System

The major contributions of the proposed system include:

- Decentralized mesh communication using BLE and Wi-Fi Direct.
- Adaptive AI-based routing for reliable message delivery.
- Priority-based rescue coordination during emergencies.
- Real-time monitoring and communication management.
- Automatic route recovery through multi-hop communication.

VI. SYSTEM MODULES

The proposed disaster response system consists of six major modules: User Communication, Mesh Networking, Adaptive Routing, Rescue Prioritization, Monitoring and Coordination, and Data Synchronization. These modules work together to support decentralized emergency communication, intelligent route selection, priority-based rescue management, real-time monitoring, and synchronization of disaster information whenever internet connectivity becomes available. The integrated framework improves communication reliability and rescue coordination during infrastructure-failure situations.

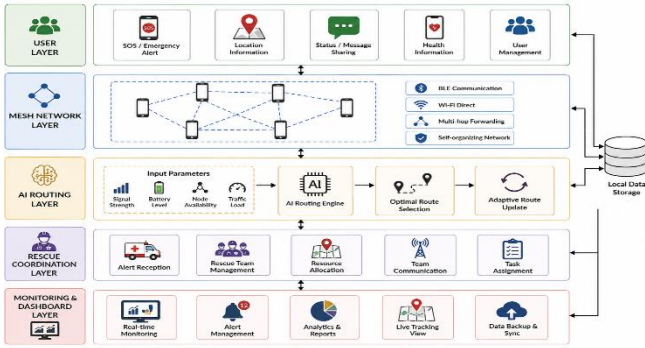


Fig. 4. Functional modules of the proposed disaster response system.

Fig. 4 shows the key functional modules that support communication, routing, rescue coordination, and monitoring.

VII. WORKING PRINCIPLE

The proposed system starts operating when a disaster situation occurs and affected individuals send emergency alerts through the mobile application. Users can provide important details such as their location, type of emergency, and required assistance. Even if internet connectivity is unavailable, the alert message can still be transmitted using Bluetooth Low Energy (BLE) and Wi-Fi Direct technologies.

Once the emergency alert is generated, the message enters the decentralized mesh network where nearby mobile devices act as mesh nodes. Each device can receive, store, and forward messages to neighboring devices through multi-hop communication. This process allows emergency messages to travel across the network until they reach rescue teams or connected monitoring systems.

The adaptive routing mechanism continuously checks network conditions such as signal strength, battery level, node availability, and communication traffic before selecting a suitable communication path. If a communication path becomes unavailable due to weak connectivity or device failure, the system automatically reroutes the message through another available node to maintain communication continuity.

At the same time, the rescue prioritization mechanism analyzes incoming emergency requests and categorizes them based on urgency and severity. High-priority requests are highlighted and processed more efficiently so that rescue teams can respond quickly to critical situations.

The monitoring dashboard provides real-time information about emergency alerts, communication activity, rescue operations, and available resources. Rescue teams can monitor ongoing situations, coordinate rescue activities, and manage resources more effectively through the dashboard interface. When internet connectivity becomes available, communication data and rescue information can also be synchronized with cloud servers for centralized monitoring and future analysis.

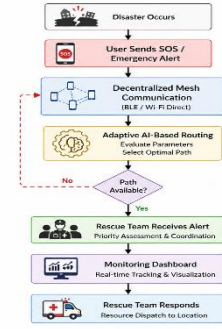


Fig. 5. Workflow of the proposed disaster response system.

Fig. 5 illustrates the overall workflow of emergency communication and rescue coordination in the proposed system.

VIII. RESULTS AND DISCUSSION

A. Experimental Setup

The proposed system was implemented using Android Studio with Bluetooth Low Energy (BLE) and Wi-Fi Direct communication support for decentralized mesh networking. Multiple mobile devices were connected within a temporary mesh network to simulate disaster communication scenarios where internet connectivity and conventional communication infrastructure were unavailable.

The performance of the system was evaluated based on parameters such as communication reliability, routing efficiency, message delivery delay, route recovery time, and rescue coordination efficiency. The adaptive routing mechanism was tested under varying network conditions including signal fluctuations, node movement, battery limitations, and communication congestion to analyze system stability and communication performance during emergency situations.

The performance of the proposed system was evaluated by comparing it with conventional centralized disaster communication systems based on communication reliability, routing efficiency, and emergency response performance.

TABLE I

PERFORMANCE COMPARISON ANALYSIS

Parameter	Existing Centralized System	Proposed System
Packet Delivery Ratio	72%	94%
Message Delivery Delay	8.6 sec	3.2 sec
Communication Reliability	Moderate	High
Network Coverage	Limited to Internet Availability	Mesh-Based Extended Coverage
Route Recovery Time	12 sec	4 sec
Emergency Message Success Rate	76%	95%
Rescue Coordination Efficiency	Medium	Improved
Communication During Infrastructure Failure	Limited	Continuous
Adaptive Routing Support	Not Available	Available
Real-Time Monitoring Support	Partial	Full Dashboard Support

The results show that the proposed system provides improved communication reliability and faster emergency response compared to conventional centralized systems. The use of decentralized mesh networking and adaptive AI-based routing reduced message delivery delay and improved successful message transmission during disaster situations. In addition, the rescue prioritization mechanism and real-time monitoring dashboard enhanced coordination efficiency and communication continuity during infrastructure-failure scenarios.

The following figures present the implementation results and key functional modules of the proposed system.

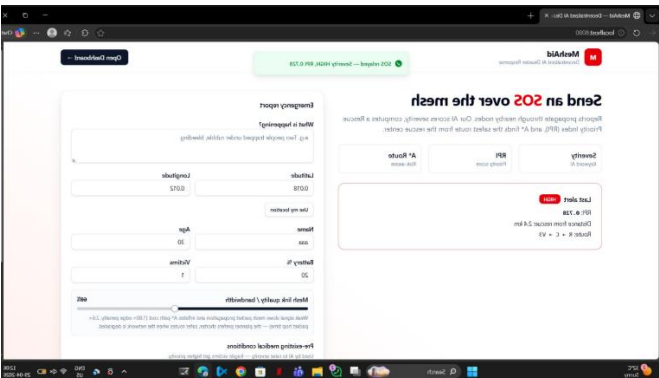


Fig. 6. Emergency alert interface of the proposed system.

Fig. 6 shows the emergency alert interface used by affected individuals to send SOS requests through the mesh network. Users can provide important details such as location, medical condition, victim count, and battery status, which help

improve rescue prioritization during emergencies.

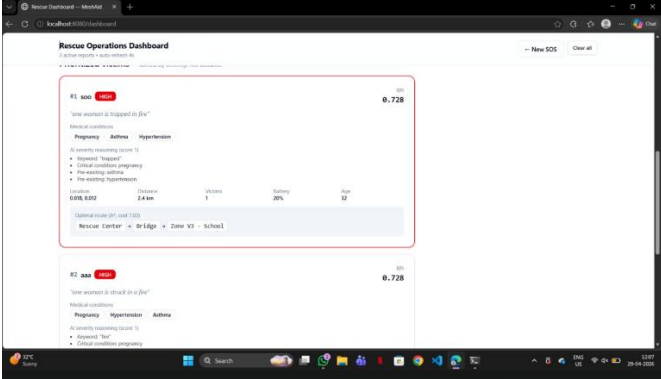


Fig. 7. Rescue operations dashboard displaying emergency alerts and priority analysis.

Fig. 7 presents the rescue operations dashboard, which displays emergency alerts along with severity analysis, rescue priority details, and routing information. This helps rescue teams identify critical cases quickly and coordinate rescue operations more efficiently.

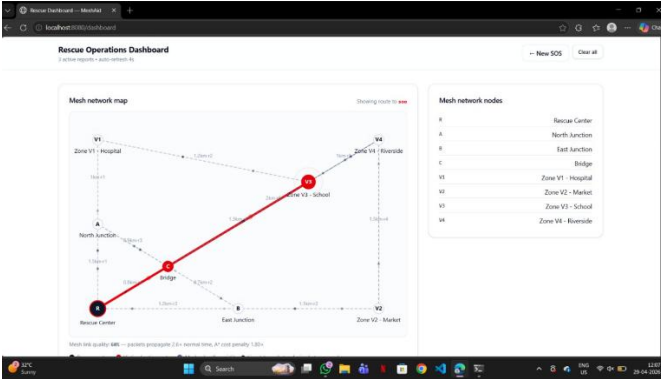


Fig. 8. Mesh network routing and communication path visualization.

Fig. 8 illustrates the mesh network routing process used in the proposed system. Emergency messages are forwarded through multiple mesh nodes using optimized communication paths to improve message delivery and communication reliability during disaster situations.

IX. CONCLUSION

This paper presented an Intelligent Decentralized Disaster Response System using Adaptive AI and Mesh Networking for emergency communication during disaster situations. The proposed system enables communication even when internet services and conventional communication infrastructure become unavailable by using Bluetooth Low Energy (BLE), Wi-Fi Direct, and decentralized mesh networking techniques.

The system combines adaptive routing, multi-hop communication, rescue prioritization, and real-time monitoring to improve communication reliability and rescue coordination during emergencies. The proposed framework also supports efficient message forwarding and helps rescue teams identify critical situations more effectively.

By integrating decentralized communication with intelligent coordination mechanisms, the proposed system provides a

reliable and scalable solution for disaster management and emergency response operations. Overall, the framework can help improve communication continuity, reduce response delays, and support better coordination during infrastructure-failure situations.

X. FUTURE SCOPE

Future enhancements to the proposed system may include:

- Integration of drone-assisted communication to improve connectivity in disaster-affected areas.
- Support for satellite-based communication in situations where communication infrastructure is completely unavailable.
- Use of advanced AI techniques for disaster prediction and intelligent rescue planning.
- Integration with government emergency response systems and smart city platforms.
- Addition of voice-based emergency alerts and multilingual communication support.

XI. REFERENCES

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